

Software Requirements Specification (SRS)

Project Name:	Online Learning Management System (LMS)
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1. Introduction

This document defines the functional and non-functional requirements of the Online Learning Management System (LMS).

The system enables students to enroll, attend classes, take tests, and view results.

Admins can manage courses, tests, students, and study materials.

2. Overall Description

The LMS is a web-based application built using React.js (Frontend),

Node.js and Express.js (Backend), and MongoDB (Database).

Authentication is handled using JWT and follows REST API architecture.

3. Functional Requirements

FR-1: The system shall allow users to register with email verification.

FR-2: The system shall authenticate users using JWT-based login.

FR-3: The system shall enforce role-based access control.

FR-4: Admin shall create, edit, and delete courses.

FR-5: Admin shall create, edit, delete, and evaluate tests.

FR-6: Students shall enroll in courses.

FR-7: Students shall attempt and submit tests.

FR-8: The system shall display results and rankings.

FR-9: Admin shall send notifications to students.

FR-10: The system shall provide doubt resolution chat functionality.

4. Non-Functional Requirements

Performance: API response time shall be under 2 seconds.

Scalability: System shall support at least 500 concurrent users.

Security: Passwords shall be encrypted using bcrypt and communication via HTTPS.

Reliability: System uptime shall be at least 99% with daily backups.

Usability: Responsive design for desktop and mobile devices.

5. System Architecture

Frontend: React.js

Backend: Node.js + Express.js

Database: MongoDB

Authentication: JWT

Deployment: Cloud-based server

6. Future Enhancements

Online payment integration
Video streaming optimization
Advanced analytics dashboard
Automated certificate generation